



Software Engineering lab - CSE 314 Lab Report

Sohoj Biniyog

Sohoj Biniyog connects you with growing businesses in Bangladesh.

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Executive Summary

This report presents the complete development and testing lifecycle of *Sohoj Biniyog*, a web-based crowdfunding platform designed to connect entrepreneurs/business owners with investors.

The platform allows business owners to create funding campaigns, while investors can securely fund those campaigns. The system supports user authentication, funding progress tracking, investment management, and automation testing using Selenium.

The project was developed using Django, HTML, CSS, JavaScript, and tested through Selenium WebDriver. The main focus areas of this report include system features, design, test plans, scripts, execution results, challenges, and future improvements.

1. Introduction

1.1 Project Overview

Sohoj Biniyog is a digital platform where business owners can raise capital from investors. The system solves real-world problems like lack of funding, trust issues, and manual communication between investors and businesses.

Key Functionalities:

- Dual role-based login (Business Owner & Investor)
- Campaign posting with title, description, target amount, and funding status
- Investors can browse campaigns and invest securely
- Real-time funding progress update
- Admin panel (Django default) for data monitoring
- Automated testing using Selenium

2. Testing Objectives

The Selenium automation suite was designed to achieve the following key objectives:

- Develop a secure crowdfunding web platform
- Allow business owners to post funding campaigns
- Enable investors to invest easily and track their investments

- Maintain database consistency using Django ORM
- Automate test cases using Selenium WebDriver

3. Tools and Testing Environment

Category	Tool / Technology	Version / Details
Automation Tool	Selenium WebDriver	Latest Stable
Programming Language	Python	3.10+
Core Libraries	selenium, WebDriver-manager, time	For browser control and script execution
Application Frontend	Html, css, javascript	
Application Backend	Django Framework	
Database	Db.sqlite3	Primary data storage
Operating System	Windows 11	Primary testing environment
IDE	VS Code	For script development
Browser Driver	Chrome driver	Chrome driver

4. Test Plan & Scenarios

A comprehensive set of test scenarios was executed, covering all major user functionalities.

Test Case ID	Description	Expected Output	Status
TC01	Business Owner Login	Redirect to dashboard	 Pass
TC02	Create Campaign	Campaign saved successfully	 Pass
TC03	Investor Login	Redirect to campaign list page	 Pass
TC04	Make an Investment	Amount updated in database	 Pass
TC05	Logout	Redirect to login/homepage	 Pass

Register_sme

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.action_chains import ActionChains
from selenium.webdriver.common.keys import Keys

# Maximize the current window that webdriver is using.
driver = webdriver.Chrome()
driver.maximize_window()

# Change the URL to the registration page
driver.get("http://127.0.0.1:8000/register/") # change if needed

# Wait for the page to load
wait = WebDriverWait(driver, 15)

# Scroll down slowly
driver.execute_script("window.scrollTo(0, document.body.scrollHeight / 2);")
time.sleep(1)
driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
print("✓ Scrolled through the page")

# Select Business Owner role
role_business = wait.until(EC.element_to_be_clickable((By.ID, "roleBusiness"))))
driver.execute_script("arguments[0].scrollIntoView(true);", role_business)
time.sleep(0.5)
driver.execute_script("arguments[0].click();", role_business)
print("✓ Selected role: Business Owner")

# Fill in registration form
username = wait.until(EC.visibility_of_element_located((By.NAME, "username"))))
email = driver.find_element(By.NAME, "email")
password = driver.find_element(By.NAME, "password")
confirm = driver.find_element(By.NAME, "confirm")

username.send_keys("business_user1234")
email.send_keys("business_user1234@example.com")
password.send_keys("TestPassword123")
confirm.send_keys("TestPassword123")
print("✓ Filled in registration details")

# Click Register
submit_btn = driver.find_element(By.ID, "submitBtn")
driver.execute_script("arguments[0].scrollIntoView(true);", submit_btn)
```

- ✓ Scrolled through the page
- ✓ Selected role: Business Owner
- ✓ Filled in registration details
- ✓ Clicked Register button

Apply for investment:

```

Click to add a breakpoint t webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time
import os

# --- Setup ---
driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 15)

# --- Step 1: Login as Business Owner ---
driver.get("http://127.0.0.1:8000/login/")

username = wait.until(EC.visibility_of_element_located((By.NAME, "username")))
password = driver.find_element(By.NAME, "password")

username.send_keys("business_user1234") # Use registered business owner credentials
password.send_keys("TestPassword123")

login_button = driver.find_element(By.CSS_SELECTOR, "button[type='submit']")
login_button.click()

# --- Step 2: Go to Apply for Investment page ---
driver.get("http://127.0.0.1:8000/apply/") # Adjust URL as needed
wait.until(EC.visibility_of_element_located((By.CSS_SELECTOR, "h2.text-center")))
print("✅ Apply for Investment page loaded")
time.sleep(1)

# --- Step 3: Fill in the form ---
business_name = driver.find_element(By.NAME, "name")
description = driver.find_element(By.NAME, "description")
location = driver.find_element(By.NAME, "location")
website = driver.find_element(By.NAME, "website")
documents = driver.find_element(By.ID, "id_documents")
terms_checkbox = driver.find_element(By.ID, "id_terms")

business_name.clear()
business_name.send_keys("My Business Test")
description.clear()

```

Post for investment:

```

from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time
import os

# --- Setup ---
driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 15)

# --- Step 1: Login as Business Owner ---
driver.get("http://127.0.0.1:8000/login/") # Adjust URL

username = wait.until(EC.visibility_of_element_located((By.NAME, "username")))
password = driver.find_element(By.NAME, "password")

username.send_keys("business_user1234") # Business owner credentials
password.send_keys("TestPassword123")

login_btn = driver.find_element(By.CSS_SELECTOR, "button[type='submit']")
login_btn.click()

# --- Step 2: Navigate to Post Campaign page ---
driver.get("http://127.0.0.1:8000/post/") # Adjust URL if needed
wait.until(EC.visibility_of_element_located((By.CSS_SELECTOR, "h2.text-center")))
print("✅ Post Campaign page loaded")
time.sleep(1)

# --- Step 3: Fill the form ---
# Select approved business
business_select = wait.until(EC.element_to_be_clickable((By.ID, "business")))
business_select.click()
business_option = business_select.find_element(By.CSS_SELECTOR, "option") # selects first approved business
business_option.click()

# Fill text fields
driver.find_element(By.NAME, "title").send_keys("My Test Campaign")
driver.find_element(By.NAME, "description").send_keys("This is a test campaign for Selenium automation.")
driver.find_element(By.NAME, "target_amount").send_keys("1000000")
driver.find_element(By.NAME, "min_investment").send_keys("1")
driver.find_element(By.NAME, "start_date").send_keys("2-10-2025")
driver.find_element(By.NAME, "end_date").send_keys("2-12-2029")

# Upload cover image (optional)
cover_input = driver.find_element(By.NAME, "cover_image")
file_path = os.path.abspath("image.jpg") # Replace with an actual test image path
cover_input.send_keys(file_path)
time.sleep(1) # wait for preview to load

```

Register_investor:

```

from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

# ✅ Setup
driver = webdriver.Chrome()
driver.maximize_window()
driver.get("http://127.0.0.1:8080/register/") # change to your local URL

wait = WebDriverWait(driver, 15)

# ✅ Scroll down slowly
driver.execute_script("window.scrollTo(0, document.body.scrollHeight / 2);")
time.sleep(1)
driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
print("✅ Scrolled through the page")

# ✅ Select Investor role
role_investor = wait.until(EC.element_to_be_clickable((By.ID, "roleInvestor")))
driver.execute_script("arguments[0].scrollIntoView(true);", role_investor)
time.sleep(0.5)
driver.execute_script("arguments[0].click();", role_investor)
print("✅ Selected role: Investor")

# ✅ Fill in registration form
username = wait.until(EC.visibility_of_element_located((By.NAME, "username")))
email = driver.find_element(By.NAME, "email")
password = driver.find_element(By.NAME, "password")
confirm = driver.find_element(By.NAME, "confirm")

username.send_keys("investor_user1234")
email.send_keys("investor_user1234@example.com")
password.send_keys("TestPassword123")
confirm.send_keys("TestPassword123")
print("✅ Filled in registration details")

# ✅ Click Register button
submit_btn = driver.find_element(By.ID, "submitBtn")
driver.execute_script("arguments[0].scrollIntoView(true);", submit_btn)
time.sleep(0.5)
driver.execute_script("arguments[0].click();", submit_btn)
print("✅ Clicked Register button")

# ✅ Optional pause to visually confirm
time.sleep(3)
driver.quit()

```

- ✅ Scrolled through the page
- ✅ Selected role: Investor
- ✅ Filled in registration details
- ✅ Clicked Register button

Investment_campaign:

```

from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

# --- Step 1: Setup ---
driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 15)

# --- Step 2: Log in as investor ---
driver.get("http://127.0.0.1:8080/login/") # adjust if your URL differs

# Fill login form
wait.until(EC.visibility_of_element_located((By.ID, "username"))).send_keys("investor_user1234")
driver.find_element(By.ID, "password").send_keys("TestPassword123")
driver.find_element(By.CSS_SELECTOR, "button[type='submit']").click()
print("✅ Logged in as Investor")

# --- Step 3: Go to campaign invest page ---
# Example: http://127.0.0.1:8080/campaign/1/invest/
driver.get("http://127.0.0.1:8080/campaign/10/invest/")
print("✅ Opened campaign investment page")

# --- Step 4: Wait for form to load ---
amount_input = wait.until(EC.visibility_of_element_located((By.ID, "amount")))

# --- Step 5: Fill investment amount ---
amount_value = "5000" # you can adjust this as needed
amount_input.clear()
amount_input.send_keys(amount_value)
print(f"✅ Entered investment amount: {amount_value}")

# --- Step 6: Submit the form ---
invest_button = wait.until(EC.element_to_be_clickable((By.XPATH, "//button[contains(text(), 'Invest Now')]")))
invest_button.click()
print("✅ Clicked 'Invest Now' button")

# --- Step 7: Wait for confirmation ---
# Wait for success alert or page change
try:
    success_message = wait.until(EC.visibility_of_element_located(
        (By.XPATH, "//div[contains(@class, 'alert-success')]")
    ))
    print("🎉 Investment submitted successfully!")
except:
    print("⚠️ No success message found. Check if investment was submitted or campaign is closed.")

time.sleep(2)
driver.quit()

```

- ✅ Logged in as Investor
- ✅ Opened campaign investment page
- ✅ Entered investment amount: 5000
- ✅ Clicked 'Invest Now' button

Campaign_details:

```
driver.maximize_window()
wait = WebDriverWait(driver, 10)

try:
    # --- Step 1: Login ---
    print("🔑 Navigating to login page...")
    driver.get(LOGIN_URL)

    # Wait for username and password fields
    username_input = wait.until(EC.visibility_of_element_located((By.NAME, "username")))
    password_input = wait.until(EC.visibility_of_element_located((By.NAME, "password")))

    # Fill in credentials
    username_input.send_keys(USERNAME)
    password_input.send_keys(PASSWORD)
    password_input.submit()
    print("🔑 Logged in as (USERNAME)")
    time.sleep(2)

    # --- Step 2: Go to Campaign Details page ---
    driver.get("http://127.0.0.1:8000/campaign/11/")
    print("✅ Opened campaign details page")
    time.sleep(1)

    # --- Step 3: Wait for campaign info to load ---
    campaign_card = wait.until(EC.visibility_of_element_located((By.CSS_SELECTOR, ".campaign-card")))
    print("✅ Campaign card loaded")
    time.sleep(1)

    # --- Step 4: Verify campaign title & business name ---
    title = driver.find_element(By.TAG_NAME, "h2").text
    assert len(title) > 0, "Campaign title missing"
    print("🟢 Campaign title: (title)")

    business = driver.find_element(By.XPATH, "//*[contains(., 'Business:')]").text
    print("🟢 Business info found: (business)")

    # --- Step 5: Progress bar and funding info ---
    progress_bar = driver.find_element(By.CSS_SELECTOR, ".progress-bar")
    percent = progress_bar.get_attribute("data-target")
    print("📊 Progress: (percent)%")

    status_text = driver.find_element(By.XPATH, "//*[contains(., 'Status:')]").text
    print("🟡 (status_text)")

    # --- Step 6: Click "Invest in this Campaign" button ---
    try:
        invest_button = wait.until(
            EC.element_to_be_clickable((By.XPATH, "//*[a[contains(text(), 'Invest in this Campaign')]]"))
    except:
        pass

    # --- Step 7: Click "Invest in this Campaign" button ---
    invest_button.click()
    print("🔗 Clicked 'Invest in this Campaign' button")
    time.sleep(2)

    # --- Step 8: Verify redirection to investment page ---
    driver.get("http://127.0.0.1:8000/invest/11/")
    print("🔗 Redirected to investment page")
    time.sleep(1)

    # --- Step 9: Verify investment form ---
    amount_field = wait.until(EC.visibility_of_element_located((By.ID, "amount")))
    print("✅ Investment form loaded (amount field visible)")

    # --- Step 10: Return to campaign details page ---
    driver.get("http://127.0.0.1:8000/campaign/11/")
    print("🔗 Returned to campaign details page")
    time.sleep(1)

    # --- Step 11: Verify campaign updates ---
    updates_section = wait.until(EC.visibility_of_element_located((By.CSS_SELECTOR, ".campaign-updates")))
    print("📄 Campaign Updates section found")

    # --- Step 12: Verify no updates posted yet ---
    no_updates_text = wait.until(EC.visibility_of_element_located((By.CSS_SELECTOR, ".no-updates")))
    print("📄 No updates posted yet – verified")

    # --- Step 13: Verify investor flow ---
    investor_flow = wait.until(EC.visibility_of_element_located((By.CSS_SELECTOR, ".investor-flow")))
    print("✅ Investor flow tested successfully!")

    # --- Step 14: Close browser ---
    driver.quit()
    print("🔌 Browser closed")
```

Portfolio:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

# --- Setup ---
driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 15)

# --- Step 1: Go to Login Page ---
driver.get("http://127.0.0.1:8000/login/") # Change base URL if needed

username_field = wait.until(EC.visibility_of_element_located((By.ID, "username")))
password_field = driver.find_element(By.ID, "password")

username_field.send_keys("investor_user1234") # your investor username
password_field.send_keys("TestPassword123") # your investor password

login_button = driver.find_element(By.XPATH, "//*[button[contains(text(), 'Login')]]")
login_button.click()
print("✅ Logged in successfully as Investor")
time.sleep(2)

# --- Step 4: Go to Investment Opportunities page ---
driver.get("http://127.0.0.1:8000/portfolio/") # change URL to your actual invest page
wait.until(EC.presence_of_element_located((By.XPATH, "//*[h2[contains(text(), 'Your Portfolio')]]")))
print("✅ Investment Opportunities page loaded")

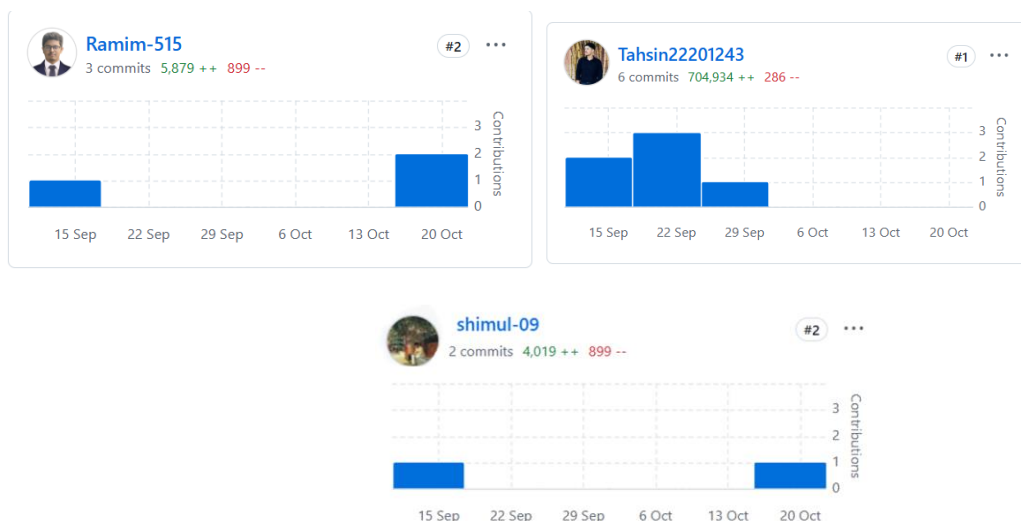
# --- Finish ---
time.sleep(2)
driver.quit()
```

6. Challenges and Solutions

Challenge	Solution
Selenium couldn't detect elements	Used Explicit Waits (WebDriverWait)
CSRF token issues	Used Django login flow & session handling
Dynamic content loading issues	Applied wait.until() with XPath/CSS
Database inconsistency during tests	Used test database / database reset
Campaign investment miscalculations	Fixed backend logic & used atomic updates
Browser driver mismatches	Used webdriver-manager for correct version
Login/Logout session problems	Managed sessions & redirects properly
UI responsiveness problems	Used CSS Flexbox/Bootstrap adjustments
Handling pop-ups/modals	Used driver.switch_to.alert() and waits
Static files not loading	Configured STATIC_ROOT and collectstatic

7. GitHub Contribution Summary

- **Repository:** <https://github.com/Tahsin22201243/SohojWeb>
- <https://github.com/Tahsin22201243/SeleniumTesting>



8. Future Recommendations

- Add Payment Gateway (bKash/Nagad)
- Add Admin Moderation System
- Add Email/SMS Notification to Investors
- Deploy Platform on Cloud (Heroku/AWS)

9. Conclusion

The automation testing for *Sohoj Biniyog* was successful and ensured that all major functionalities of the platform—such as user authentication, campaign creation, and investment processes—performed as expected. Selenium-based test scripts helped validate both frontend interactions and backend logic, ensuring a smooth and reliable user experience. The integration between the Django backend and the HTML/CSS/JavaScript frontend was tested thoroughly and proven to be stable. Automated testing not only reduced manual effort but also helped detect errors early, ensured bug-free releases, and improved overall system performance. Overall, automation testing has become an essential part of the development process for *Sohoj Biniyog*, enhancing accuracy, increasing efficiency, and boosting confidence in the final product.

10. References

1. Selenium Official Documentation. <https://www.selenium.dev>
2. Django Framework Documentation. <https://www.djangoproject.com/>
3. WebDriver Manager for Python. <https://pypi.org/project/webdriver-manager/>
4. Python Official Documentation. <https://docs.python.org/3/>

Selenium Testing Video:

[https://drive.google.com/file/d/1cHsG5V5Nlc5bpwUidQwKFSMql-U0VMwC/view?usp=drive link](https://drive.google.com/file/d/1cHsG5V5Nlc5bpwUidQwKFSMql-U0VMwC/view?usp=drive_link)